ClubSync: Comprehensive Project Report & Engineering Dossier

Target Institution: SCTR's Pune Institute of Computer Technology (PICT), Pune | **Date:** October 2026
Platform Scope: Unified Collegiate Society & Event Operations Architecture

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1. Executive Summary & Problem Context

Collegiate student societies and technical clubs at premier engineering institutions like PICT (e.g. IEEE, ACM, CSI, EDC, Robocon) face acute operational fragmentation across disconnected tools:

- **Gate Bottlenecks & Proxy Attendance:** Registrations via Google Forms cause long queues, proxy check-ins, and manual ticket verification failures.
- **Treasury & Audit Risk:** Student payments bounce between personal UPI accounts, forcing officers to manually trace bank UTRs with zero gymkhana audit readiness.
- **Fragmented Sponsorship Outreach:** Corporate sponsorships get lost in personal WhatsApp threads, leading to dropped brand deliverables and lost funding.
- **Governance Breakdown:** Constitutional by-laws and annual officer elections are conducted informally without verifiable quorum or democratic supermajority tracking.
- **Academic Compliance:** Manual Dean On-Duty (OD/DL) leave letters and paper certificates are slow to approve and prone to fraudulent tampering.

ClubSync solves this through a unified, high-concurrency collegiate platform engineered specifically for engineering campus operations.

2. System Architecture & Engineering Highlights

ClubSync employs a modern reactive architecture designed for zero latency and enterprise-grade security compliance:

Tier / Layer	Technology Choice	Engineering Capabilities & Guarantees
Frontend Client	React 18, Vite, Tailwind CSS, Framer Motion	Mobile-first tactile UI with dark mode, sub-100ms transitions, and PWA service worker caching for offline auditorium reliability.
Backend Engine	Node.js 20, Express REST API, Helmet, CORS	Hardened endpoints with Helmet security headers, compression, rate-limiting, and strict BOLA/IDOR authorization middleware.
Database & Storage	Neon Serverless PostgreSQL (pg-pool)	18+ normalized tables, composite foreign keys, ACID compliance, and 100% parameterized SQL query defense ($1, $2) with zero injection risk.
Real-Time & AI	Discord/Telegram Webhooks, Groq Llama 3	Multi-channel broadcast syndicator with health telemetry, plus sub-second AI assistant for constitutional queries and meeting summarization.

3. Workload Distribution Matrix (6 Team Members)

Workload is distributed across 6 specialized functional domains, with the Team Leader commanding architecture and core security (35% weighted contribution) and 5 members commanding specialized subsystems (13% weighted contribution each):

Role / Member	Subsystem Scope	Key Architectural Modules Built	Tech Stack
Member 1 (Team Leader)	Lead Architect & Database Lead *(35% Weight)*	• System Architecture & PRD formulation • Neon Serverless PostgreSQL relational schema (18+ tables) • Express REST API foundation & Helmet security headers • JWT auth lifecycle & BOLA/IDOR protection • 100% Parameterized SQL injection defense	Node.js 20, Express, PostgreSQL, Neon DB, JWT, Helmet

Member 2	Event Operations & Hardware *(13% Weight)*	• Event Day Desk Kiosk Mode (EventKioskModal.jsx) • Continuous USB/Bluetooth scanner wedge listener • Instant Thermal Receipt & Badge Printing (80mm) • 15-second Spot Walk-In gate registration pipeline • Anti-screenshot digital passes with SHA tokens	Canvas API, Web Audio API, DOM Print, React Hooks
Member 3	Financial Systems & CRM Lead *(13% Weight)*	• YouPI Zero-Fee Collegiate Payment Gateway • Direct UPI deep-linking (GPay/PhonePe/Paytm) • Tamper-proof collegiate treasury ledger & e-receipts • Corporate Sponsorship & Outreach CRM Kanban Board • 1-Click Gymkhana Audit CSV export engine	React State, CSV Parsing, UPI Protocol, Kanban UX
Member 4	Governance & Compliance *(13% Weight)*	• Constitutional By-Law Amendment Portal • Side-by-side constitutional red/green diff box • 66.6% supermajority & quorum validation engine • Senior-Junior Peer Mentorship Pairing Hub • Automated Dean On-Duty (OD/DL) Leave Generator	Text Diffing, Voting Logic, jsPDF, Math Engine
Member 5	Communications & AI Lead *(13% Weight)*	• Multi-Channel Discord & Telegram Webhook Syndicate • Automated color-coded embed generation & blast blaster • Webhook dispatch health logs (club_webhook_logs) • Slido-style Keynote Live Stage Q&A; with upvoting • Groq Llama 3 AI for constitutional RAG & minutes summary	Discord/Telegram APIs, Groq AI SDK, Polling
Member 6	UI/UX & Public Credentials *(13% Weight)*	• Responsive UI/UX Design System with Tailwind CSS • Progressive Web App (PWA) Workbox precaching • Public Certificate Anti-Fraud Verification (/verify/:id) • 1-Click 'Add to LinkedIn Profile' credential exporter • Robotics Lab Equipment Inventory Checkout Tracker	Tailwind CSS, Vite PWA, Framer Motion, Workbox

4. Technical Viva Defense & Evaluator Rehearsal Q&A;

Q1: How do you prevent double-voting in an election or referendum?

In our PostgreSQL schema, the `club_resolution_votes` and `election_votes` tables enforce composite unique constraints: `UNIQUE(resolution_id, voter_id)` and `UNIQUE(election_id, voter_id)`. Any duplicate submission is rejected at the database engine level with a 409 Conflict error, guaranteeing mathematical vote integrity.

Q2: What happens if campus Wi-Fi drops during gate check-in at the auditorium?

ClubSync operates as a Service-Worker powered Progressive Web App (PWA). Kiosk scanner assets and cached registration rosters reside in browser storage. The scanner wedge continues listening to incoming hardware streams and synchronizes state upon network re-establishment.

Q3: How do you eliminate SQL Injection and Broken Object-Level Authorization (BOLA)?

100% of our database queries strictly use pg parameterized tokens ($1, $2) with zero string concatenation. On the authorization layer, `requireOfficer` and `requireAuth` middlewares verify ownership at the controller level: students cannot inspect attendee rosters or other members' payment receipts by altering URL parameters.

Q4: Why not simply use Eventbrite or Google Forms?

Eventbrite takes 2-3% commission fees and lacks Indian UPI integration, AICTE credit tracking, and collegiate Dean On-Duty letter generation. Google Forms allows proxy attendance, lacks barcode check-in, provides zero financial reconciliation, and cannot drive live Discord syndication or thermal badges.